
INFORMATION ANALYSIS

Information Analysis in Risk Management of BP (Using the Example of Metallurgical Enterprises)

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Abstract—To unify the understanding of and improvements to management efficiency, the conceptual apparatus of information support for risk management of BP is clarified, and methods of analysis and assessment of risk in the information environment are explored, using the example of a metallurgical company. The development of economic and mathematical support in calculating a quantitative measure of risk is presented, and a matrix for calculating the correction weight coefficient W_i for the determination if this measure is proposed, which allows us to take into account the duration and importance of identified risk factors, rank risks, and make informed decisions based on a more complete and detailed assessment of them.

Keywords: information, BP, risks, management, metallurgy, information analysis methods, correction factors

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INTRODUCTION

The importance of scientific and technical information is growing rapidly. The skillful use of digital technologies in production is a condition for the effective risk management of modern business processes (BP) in general and in the metallurgical industry in particular. This article examines the impact of information on the speed, flexibility and economic efficiency of BP in the metallurgical industry. The Russian vertically integrated metallurgical holding PAO Severstal was chosen as the object of study.

This work is relevant because, in the twenty-first century, along with traditional social, technical, technological, and economic risks, risks caused by the geopolitical crisis and the challenges of the fourth industrial revolution have come to the fore. Information is now a key component in BP management. Breakthrough information technologies enable businesses to operate and reconfigure themselves much more rapidly and efficiently. Forecasts indicate that by 2030 the economic effect from the use of digital technologies will amount to tens of trillions of dollars [1]. At the same time, the number of cybercrimes at different levels is increasing worldwide; new types, numbers, and sizes of business risks are multiplying; and the cost of their identification and assessment as well as the reduction of their negative consequences are increasing. For this reason, the study of risks in the context of digitalization of production management is becoming an increasingly important area in scientific and practical research and development.

The results of an international survey of risk managers leading enterprise-wide risk management system (ERMS) the Fortune 1000 companies list their greatest concerns as relating to the following types of risks: cybersecurity (45%), customer fraud (30%), and confidentiality of customer and employee data (28%). Companies are using new methods of risk management, including dynamic risk monitoring (30%), new technical solutions for risk management (25%), and data analytics (24%) [2]. The growth of cyber threats has led to the tendency to increase the number and equipment of security services. The relevance of using multi-factor authentication to protect information is growing. According to the Gartner forecast, the main trend in the development of business by 2025 will be the digitalization of BP and strengthening of the information component of ERMS [3].

To develop and improve the BP of metallurgical production, it is necessary to keep up with the times through intensifying the import substitution of digital support for ERMS companies. In March 2022 the company Systems, Applications and Products in Data Processing (SAP) halted all new sales in the Russian Federation and the Republic of Belarus and began to curtail the provision of cloud services, while all BPs of PAO Severstal were built on the use of foreign software, including SAP systems (ERP, S4 Hana, Analytic Cloud (SAC), and others), Manufacturing Execution System (MES), QMET (for remote warehouses and so on), BPM-systems, and foreign providers. PAO Severstal received grant support from the Skolkovo Foundation for the development of two large projects in

import substitution in foreign software in key sectors of the Russian economy; the total amount of these grants exceeded 1 billion rubles. The support measure is provided for by the national program “Digital Economy of the Russian Federation.” At this time, important resources from the company’s profits are allocated to the development of its own software. In recent years, the digitalization of the production of a process management system in metallurgy (MES Metallurgy) and the creation of the planning optimization system Severstal has already allocated more than 1.6 billion rubles of its own funds and expects that the solutions will be ready for implementation by 2026. The total investments in information technology and digital projects will amount to \$76 million, indicating that metallurgists’ own IT investments are tens and sometimes hundreds of times greater than government investments [4].

Scientific and practical justification of the ways and means of increasing the efficiency of digitalization ERMS should not lag behind the development of BP. The market is seeking opportunities to adapt to new conditions, help metallurgical enterprises maintain the functionality of their existing IT infrastructure, and ensure increased production efficiency through the technological support and risk reduction of BPs.

CONCEPTUAL APPARATUS FOR INFORMATION ANALYSIS OF BUSINESS PROCESSES

Information support for risk management in BP is a complex, dynamic phenomenon that requires uniformity and mutual understanding in the writing and deciphering of text and other information search data at all levels of interaction of an ERMS, so we should begin with an analysis of the theoretical characteristics of the conceptual block.

There are many interpretations of the essence of the category of information, and rapid changes in its content make it difficult to form a universal, generally accepted definition. Thus, A.G. Khanzhin and A.A. Cojocararu consider information to be any information perceived as such by the intellect [5]. The role of an institutional basis of information theory is played by a mathematical theory of communication [6], which, however, cannot not take into account evolutionary changes in the essence of information. In the anthropocentric concept (F.N. Tsidrya, V.G. Afanasyev, and Yu.A. Schrader) information is inextricably linked to the subject of its perception [7]. A physical view of information suggests that the transfer of information requires both an information carrier and a recipient. This view was defended by R. Landauer, who argued that information has physical properties, spreads from one point to another, and is transmitted through interactions between the points of departure and reception of information. The assessment of

information, in turn, is interconnected with the individual and his intelligence [8].

There are different points of view on the nature of information. Its essence has been studied by many domestic and international scientists in different fields of knowledge: mathematics, economics, philosophy, physics, cybernetics, and so on (L. Gurvits, A.N. Kolmogorov, R.M. Nizhegorodtsev, I.A. Achkurin, K. Shannon, R. Landauer, O. Lombardi, and others).

Information is prioritized in decision making. According to L. Hurwitz, the inefficiency of a market economy is due to incomplete or erroneous information regarding its subjects. Thus, information is a factor in production, occupying a key place in the information economy [9]. According to O. Lombardi and others, information is a commodity that, if it is received correctly, can provide knowledge and therefore income [10].

The development of information and network technologies, the popularization of remote execution of certain production and management operations, introduction of the Internet of Things, and related changes have caused a significant increase in the volume of information used, leading to an increase in the number, types, and level of risks for enterprises that are actively integrated into the digital environment [11].

Taking into account that the object of this study is the effectiveness of the risk management of BP of metallurgical enterprises and the subject is a person, we believe that when clarifying the essence of information, it will be more expedient to take the functional approach of P.V. Kopnina, V.S. Tyukhtina, N.N. Moiseeva, D.I. Dubrovsky [12], and a number of other scientists as a basis. According to their interpretation, information is necessary for effective management of the process (object). Management is based on the analysis and assessment of ongoing changes in BP elements [13]. Information is also considered as a system of data existing in the external and internal environment of a certain object, as recognized by individuals, making it possible to control this object [14]. Under the influence of information, transformational transformations of BP occur [15].

Next, we should clarify the concepts of BP, which in this case are the object of ERMS management and its information component. A.V. Scheer provided the generally accepted and most established definition of BP for theory and practice: “it is a connected set of repeatable actions that transform input material and/or information into a final product in accordance with pre-established rules” [16]. O.N. Rodtsevich defines the goal of BP as creating economic value for the organization through meeting the needs of internal and external clients for resources, considering such processes as integral elements of this organization [17]. According to P.V. Kutelev BP consists of grouped and interacting business functions to create a product or groups of products that are targeted at a specific con-

sumer. This does not specify which consumer we are discussing—final or intermediate—but such functions are specified through the relationship based on the creation of a specific product [18]. M. Hammer and J. Champi define BP as a complex of activities wherein certain resources are used at the input, and at the output, a product is formed that is in demand by the consumer [19]. N.R. Kelchevskaya specifies the innovative essence of BP of industrial enterprises [20].

In our opinion, the methodological basis for the effective informatization of risk management of BP is an integrated approach, the elements of which are the process, system, and situational approaches. In this case, corporate management is considered to be management of the BP complex [21]. The introduction of a process approach in metallurgical enterprises is intended to increase efficiency through the elimination of duplicate operations. In practice, an integrated approach in the late 1990s enabled the reduction of the level of multitasking and to increase the efficiency of management of multi-industry corporations (General Electric, Toyota, and Chrysler).

The essence of the concept of BP requires addition and clarification, due to an increase in the volume of information that arises in the framework of the preparation and implementation of BP and their individual operations, as well as a growing number and increasing complexity of relationships and interdependencies between BPs. This is especially typical for large, vertically integrated metallurgical companies (PAO Severstal, PAO NLMK, OOO Evraz, and others).

To identify an independent BP at a metallurgical enterprise, from the point of view of its essence, it is necessary to determine at least five of its effective characteristics. First, these are measurable targets (KPI). The lack of generated key performance indicators indicates that these activities are not interrelated and cannot be combined into one BP. A system of criteria must be formed to characterize the final result of the BP (profit, revenue, EBITDA (profit before interest on loans, income tax, and depreciation), costs, project timing, etc.). Second, this is a specific added value. BPs create a basis for the owner of the enterprise to make a profit. If the performance of the BP will not create added value directly (operational) or indirectly (infrastructure, management), then either it should be merged with another BP or the list of its key indicators should be updated. Third, it is necessary to specify the manager (owner) of the BP, the responsible person who has with certain resources available (material, human, temporal, etc.) must be specified. Fourth is the identification of BP boundaries (definitions of input and output), established using organizational resources (input) and the creation of added value (output) for an external or intermediate consumer. Finally, BP flexibility should be ensured through manageability and monitoring. This is necessary to respond in a timely manner to changes in the external or internal environ-

ment of BP. The lack of this flexibility leads to wasted time and reduced added value.

Having supplemented the economic essence of BP with its key characteristics, we clarify the essence of BP risks. The national standard of the Russian Federation “Risk management. Terms and Definitions” defines risks as a result of the influence of uncertainty on the achievement of specific goals [22]. We believe that it is possible to understand this as the amount of deviation in the resulting indicators at any stage of BP from standard or calculated indicators of quality, time, efficiency, and so on. Therefore, we propose to understand BP risks in accordance with the definition of F.Ya. Legotina and A.A. Shishkina [23] as “the threat of loss of control over the amount of income (expenses) in the process of implementing a project or business operation.” In this context, we consider BP risks to be the likelihood of losses or reductions in profit due to the occurrence of an unfavorable situation at individual stages of BP.

To clarify deviations of the results of a BP from the standard, you should understand what the effectiveness of the given BP. For example, companies use BP algorithms, including the name of the process, content, purpose, functions, its place in a number of other processes, and the order of execution. However, this algorithm does not specify the planned indicators efficiency. At each stage of BP it is necessary to measure, analyze, adjust and refine its performance indicators, which are the functions of ERMS. Information analysis and the computerization of management decision-making play a crucial role here, as any solution must be closely integrated into existing industrial enterprise management information systems [10].

INFORMATION AND RISK MANAGEMENT IN METALLURGY

By the beginning of the 21st century, huge amounts of data had accumulated in the business environment that require new collection and processing technologies. Technological innovations, artificial intelligence, blockchain, robotization, cloud platforms, machine learning, and the Internet of Things are actively developing and being used in the metallurgical industry. The use of innovative technologies leads to a constant transformation of forms of labor in metal production, especially in management systems and building BP taking into account these changes. Network flows and digital forms of communication accelerate everything that happens in the industry and beyond, significantly accelerating risk management in BP. Metallurgical companies actively use digital technologies.

The cloud environment is the most promising direction for the development of information technology in the ERMS. More and more, not only the management but also the computing workloads of metallurgical enterprises are moving to the public cloud as

part of technological Big Data to timely receive the information necessary for making decisions in the RMS in a timely way [24]. Risk helps the system become “antifragile,” as many new solutions must be found in a big data system. By antifragile, N. Taleb understands the ability of a system to withstand unpredictable events and become stronger [25].

Risk management can be optimized using five main analytical tools: (1) the identification of risks arising in the internal and external environment; (2) the assessment, ranking, and prioritization of risks that need to be managed to ensure that the critical areas of BP receive special attention; (3) risk response and mitigation: to model what-if scenarios. Risk modeling methods can be adopted through the integration of data elements, which help select the most optimal response strategy in the RMS; (4) risk monitoring and assessment, where identifying key indicators for each risk is important to measure trends and changes in data parameters associated with risks; and (5) permanent risk reporting in real to effectively implement by embedding the life cycle of an ERMS into a technology platform [26].

In all of these tools, information occupies a key position: effective risk management largely depends on its objectivity, completeness, and timeliness.

Another modern management approach to Big Data technology is a “data factory” or architecture for working with information, which allows companies deliver data where it is needed most and securely manage it, making it easier to implement consistent policies across the application ecosystem and users. In this approach, more data sets and digital platforms can be easily added by connecting new resources, whether external or internal, and new services can be added or removed as needed without interruption. This improves data quality, provenance and process traceability, and reduces the risk of using poor quality data in strategic decision making [27].

Hyperautomation is another information trend in ERMS BP, including two main blocks of technologies: (1) Robotic process automation (RPA), using the metaphorical software of robots (bots) or artificial intelligence workers. Most often, this technology is implemented as an add-on on top of an existing software landscape, performing a set of actions in a fully automatic mode with information processed by the operator manually ahead of time. Among the advantages of an RPA is that it saves time from routine tasks, allows the business to free up (reduce) operational time and human resources, reduce operating costs, and increase the speed of task completion; (2) Robotic desktop automation, a type of RPA, implies partial automation of the process in the form of separate, often specific, operations. It allows the time required to perform previously manual routine operations such as data extraction, transfer, integration, and conversion to be significantly reduced; transactions in production management sys-

tems (ERP), writing letters, working with S.M.\I.M., data analysis, report generation, and so on.

These information technologies are in active use in metallurgy at PAO Severstal, where the risk management and internal control system is made up of a set of interrelated components, the structure of which corresponds to the generally accepted methodologies COSO ERM and COSO IC-IF.

For COSO ERM, “Organizational risk management. Integration with strategy and performance efficiency,” and for COSO IC-IF, “Internal control. Integrated Model” are methodological documents in the field of risk management and internal control. Due to scaling at Severstal-CES (Unified Service Center) there the need arose to develop a single platform for recording and processing all information on projects, as well as organizing an environment for coordinated interaction between teams involved in working with customers. The tool for implementing this project was CRM-system Sales Creatio from Terrasoft.

To monitor BP, PAO Severstal uses the tool Power Business Intelligence (Power BI) for dashboarding. Before dashboards (a special control panel that displays key metrics and indicators, such as the number of sales, etc.) for conducting analytics, we used Excel. SAC is a cloud solution that combines the capabilities Power BI supplemented by predictive analytics and planning; the tool is suitable for creating dashboards and other data visualization methods. First of all, it is intended for the system SAP, but can also work with other data sources. Because at Severstal, a significant part of BP is automated using SAP, SAC fits into the company’s infrastructure.

Currently, communication process in the Severstal company is established using tools of electronic data exchange and document flow, e-mail information, the Severstal Market online store, and other digital services. The company prioritizes risks to formulate a strategy to respond to them and allocate resources to manage them. The choice of strategy for the response to risks is determined taking into account business conditions and the ratio of benefits and costs. Let us consider the impact of information technology on the effectiveness of risk management at PAO Severstal (Table 1).

The modern practice of uncertainty minimization adopts many different approaches, techniques, and private tools [28]. In Table 2 we will consider the most technologically advanced modern methods of risk analysis and assessment.

It should be noted that to assess the risks of the BP of PAO Severstal, along with the classical method SWOT-analysis, the most effective method to use Business Impact Analysis (BIA), as the totality of the company’s information technologies makes it possible to collect complete data on all areas of activity. BIA requires a large amount of input data and is suitable for large enterprises, such as PAO Severstal, which is a vertically integrated company and includes in its

Table 1. Role of information in risk management of PAO Severstal

Applied information technologies	Business process	Impact on risk management
Microsoft Office Solution Accelerator for Sarbanes Oxley	Risk management	The cycle of risk management processes is automated, the ability to track control procedures and generate operational and analytical reports is provided.
Avanpost IDM is a unified system for centralized management and access control in the central office and in the holding's subsidiaries	Transformations of basic IT services	Replacement of the SAP GRC system and other solutions whose functionality falls under the responsibility of IDM. Managing user authentication in corporate resources and cloud services allows you to minimize the risk of leakage of confidential and personal data and loss of access to user accounts.
SAP ERP—HumanCapitalManagement	Personnel management	Organizational management processes, operational reporting, corporate training, personnel administration, time data management, payroll, services for employees and managers, and employee interaction center processes have been automated. Transparency and efficient use of labor resources are ensured through the unification of processes and standards.
SAP S/4HANA is a digital core that takes into account production operations, resources, financial results and company assets	Operating activities	The technology platform increased system performance and increased the speed of processes. It became possible to keep track of financial results not by individual business units, but by streams (upstream and downstream), which reflects the existing organization of the company. The platform also made it possible to prepare end-to-end reporting for the entire group of companies.
BPM systems	Operating activities	Enables the optimization and acceleration of the processes of sales, marketing, and service and operational processes of the organization, as well as monitoring compliance with deadlines, budgets, and other project restrictions based on real data, guarantees the project's readiness for the occurrence of risks and thorough documentation of risks.
CRM system Sales Creation from Terrasoft	Control procedures	A single platform for recording and processing all information about projects, as well as organizing an environment for coordinated interaction between teams involved in working with customers. The program is aimed at preventing, identifying and correcting errors and distortions of information in the accounting system.
MES systems: QMET—metal tracking system	Operating activities	Optimization of warehouse operation, improvement of product delivery discipline, reduction of consumption coefficient by eliminating unnecessary movement of metal and associated injuries. The loading and cutting plan is carried out with greater accuracy; in general, warehouse capacity and turnover increased.
MES systems: Remote warehouses (RS)	Operating activities	Managing the movement of materials from the production facilities of PAO Severstal to tracking the receipt of materials at remote warehouses (reception and shipment of materials). The program allows the current state of availability of metal products to be displayed in the warehouse and reduces the risk of the formation of excess usable metal and eliminates cluttering of warehouses.
RBP. Lean technologies	Operating activities	The company has implemented lean manufacturing tools and technologies: Agile, Kaizen, 5C, etc. Projects are being implemented based on lean manufacturing to improve process efficiency, which include optimization of production processes, continuous improvement of product quality, and cost reduction.

Table 2. Analysis of information methods for assessing risks of BP

Methods	Characteristic	Weak points
BIA	Analysis of how key risks can affect the functioning of the organization. Identifies and quantifies opportunities to manage them. Risk assessment method allowing you to study different types of negative events affecting key BPs. Assessed by a panel of experts based on initial BP information.	1. The competence of participants in working groups and interviews may be insufficient. 2. The characteristics of a group's work can directly influence the results of BP analysis. Subjective assessments of RTO (recovery time) requirements are possible—either too optimistic or too pessimistic. Achieving an adequate level of understanding of an organization's BP and its activities is very difficult.
FMECA	Failure mode, effects and criticality analysis (FMECA) is a systematic approach to identifying potential failures, assessing consequences, and identifying critical system components. Allows the identification of possible causes of failures and the assessment of their impact on operation business systems and develop measures to eliminate or minimize risks. FMECA is used in a variety of industries, including manufacturing, aviation, medical and others, to ensure the safety and reliability of systems and processes.	1. Complexity and cost: conducting an FMECA requires significant resources, time and expertise. 2. Data limitations: conducting FMECA requires access to sufficient data about the system, its components, and its processes, which may be difficult in some cases. 3. Lack of accuracy: estimates of the probability of failure and its consequences may not be accurate enough due to lack of data or incomplete information. 4. Need for continuous updating: systems and processes can change over time, requiring FMECA to be constantly updated to ensure it is current and effective.
HRA	Assessing operator reliability concerns the impact of personnel on the functioning of a business system and can be used to assess the impact of operator errors (the human factor) on it. This method of assessing employee reliability is aimed at identifying errors that negatively affect the reliability of the organization. HRA also identifies the methods that cause such errors and other failures can be corrected by the employee.	1. The complexity and variety of types and parameters of operator behavior creates significant difficulties in determining simple failure modes and assessing their probability. 2. It is impossible to describe many operator actions using the concepts of working and unworkable states. 3. The HRA method is difficult to apply in situations with partial failures or failures due to inappropriate decisions made.
LOPA	Layers of protection analysis (LOPA) method is a risk analysis method used to determine the necessary levels of protection and control that must be applied to prevent or minimize potential hazards. It is based on the idea that security systems can be divided into layers of protection, each of which represents a separate security measure. LOPA allows you to evaluate the effectiveness of each layer and determine what additional measures are needed to achieve the required level of security.	1. Limited consideration of all possible risks and safety measures. It is possible that some risks may be missed if they are not taken into account in the analysis. 2. The LOPA method can be subjective because estimates of the effectiveness of each layer of protection may depend on the personal preferences and experience of the experts conducting the analysis. 3. Can be expensive and require significant time and resources to conduct the analysis.
Markov method	The Markov method is a method of risk analysis based on a mathematical model that enables the estimation the probability of occurrence of certain events depending on previous events. It allows all possible risks and safety measures to be taken into account, and it is also less subjective than the LOPA method. However, it can be expensive and require highly qualified specialists to use it.	1. A large amount of data is required to build a mathematical model, which is difficult in the absence of sufficient statistical information. 2. The model may be sensitive to the choice of initial conditions and parameters, which may lead to inaccurate results. 3. Does not take the influence of external factors on the occurrence of risks into account. 4. When operating conditions of the B-system change, it is necessary to rebuild the model, which requires additional time and resources. 5. The model does not take the human factor into account.

Table 3. Weighting coefficient calculation matrix (W_i) income value for event “ n ”

Duration of action	Degree of influence of risk on the business system		
	low	average	high
From 0 to 3 years	0.1	0.175	0.25
From 3 to 5 years	0.125	0.2	0.275
More than 5 years	0.15	0.225	0.3

structure a complex of enterprises according to the stages of the technological process.

BIA acts as a strategic planning tool, as it can be used to develop scenarios in the event of failures in the operation of core BP and business operations. The results of business impact analysis within the framework of the BIA can help a company effectively mitigate the consequences of critical risks [27].

However, the use of individual methods for qualitative risk assessment is insufficient. To comprehensively assess and monitor of risks, they must be assessed and quantified. Thus, each of the outcomes developed during BIA scenario analysis can be complemented with sensitivity analysis. The method of risk sensitivity analysis is among the main methods of quantitative analysis. Its main orientation is to analyze the degree of variability of performance indicators in relation to varying project parameters. The more actively that the economic efficiency indicators of an investment project react to changes in input quantities, the greater the exposure of the project to corresponding risk [29].

Some experts have proposed updating the quantitative risk assessment tool [30]—a method for determining the possible amount of profit or loss when making a management decision. The mathematical expectation of the corresponding damage (or expected income in speculative risk models) is taken as the risk measure R :

$$R = \sum_{i=1}^n P_i X_i, \quad (1)$$

where R is quantitative measure of risk, namely, average risk, expressed in the same indicators as damage (income);

P_i is the probability of receiving damage (income) of size X_i as a result of the occurrence of any event (group of events);

X_i is the amount of damage (income), expressed in appropriate indicators (in economics, as a rule, in value terms);

n is the number of possible damage (income) options that may occur, including zero damage.

This is the classic formula for determining risk measures. In the context of an increase in the level of risks in

the implementation of the company's BP, caused by exogenous changes, the need to increase the accuracy of the assessment of the analyzed risks increases. We propose a weighting correction factor (W_i) for analysis of the amount of income for an individual event or factor, due to its importance or the duration of its action.

The content of other indicators remains unchanged. The weighting coefficient value (W_i) is determined based on the expert assessment of the level of influence that a specific event will have on the company's business system. Henceforth W_i adjusted and adapted depending on the specific BP and its features. The correction factor (W_i) is reflected in the matrix (Table 3), in which the values in the cells represent weighting factors that take into account the importance or duration of the relevant risk.

For example, if the event lasts up to 3 years and the degree of impact of the risk is also low (having a small impact on the project), then the weighting coefficient will be 0.1, and the amount of income will be reduced by only 10%. Using this matrix, the overall risk of a project can be more accurately assessed, and appropriate measures can be taken to manage it.

Thus, the modified formula for the quantitative measure of risk will look like this:

$$R = (X_1 P_1 W_1) + \dots + (X_n P_n W_n) = \sum_{i=1}^n X_i P_i W_i, \quad (2)$$

where X_i and X_n are values of financial indicators associated with the i th and n th events or risk factors;

P_i and P_n are probabilities of occurrence of the i th and j th events or risk factors;

W_i and W_n are weighting coefficients, taking into account the duration or degree of influence of the risk of the i th and n th events.

The use of the proposed correction factor (W_i) will make it possible to make more informed decisions based on the objective consideration of a set of risk level factors (duration of the event and importance and degree of influence on the company's business system). Each method has its own advantages and disadvantages and allows the comprehensive analysis of the risks of BP from different angles.

CONCLUSIONS

Informatization forms the basis for increasing the efficiency of the risk management of BP in metallurgy. PAO Severstal is intended to create a transparent ERMS, developing the hierarchy and comprehensiveness of BP information analysis, taking online risk analysis into account that would allow the company to respond to new challenges in a comprehensive and timely manner. By automating risk management processes and improving data availability, analytical capabilities, communication, and monitoring, it is possible to reduce the likelihood of errors, increase operational efficiency, and create more informed decisions.

Clarified concepts of information, BP, and risks create the opportunity to achieve uniformity in monitoring, analyzing, assessing, adjusting, and improving the risk management of BP of metallurgical enterprises from the point of view of a comprehensive study of problems and identifying means of improving BP.

We propose to solve the issue of increasing the efficiency of information analysis in the risk management of BP of metallurgical enterprises on the basis of the integrated use of positive aspects and taking into account the possible negative consequences of the studied and analyzed qualitative and quantitative methods of analysis and risk assessment, such as BIA and FMECA. In our opinion, it is also advisable to carry out industrial adaptation of modern information technologies (MOSASO, CRM-system Sales Creatio, MES-systems, and others). Many large metallurgical companies have successfully implemented processes of import substitution for corporate information systems (SAP, Oracle, and so on) with minimal losses for main and auxiliary production.

The formulas we propose for quantitative assessment of risks and the factors influencing them will allow us to process ever-increasing amounts of information with minimal expenditure of resources and time. The transformation by metallurgical companies of a general ERMS based on digitalization and the integrated use of qualitative and quantitative methods for assessing the risks of individual BP is intended to reduce the overall level of risk and improving the constituent elements of the business system of metallurgical enterprises.

Under modern conditions, information technologies have a direct impact on the quality of management decisions and ensuring the reduction of risks of BP in metallurgy. This influence begins with pre-project information analysis and risk assessment, continues with the planning and forecasting of risk management methods, and permanent monitoring with adjustment of risks and their uncertainty. In the context of a deteriorating external economic environment and growing internal problems, the successful operation of an enterprise depends on the timely adjustment of established business strategies in the event of changes in the external and internal environments.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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